

with previous-generation Wi-Fi 5 or Wi-Fi 6 standard to help achieve low power.

Low power is usually achieved by keeping the wireless device in sleeping mode most of the time, but that increases latency, which makes response time of the system very slow. In many applications this is not possible because faster response time, data rate and low power are also needed.

The goal is to achieve both low power and low latency. This is achieved by low-power wake-up receiver (LPWUR). It works in conjunction with the normal WLAN transceiver (TRx). The basic function of wake-up receiver is to sniff the air for wake-up packets. Once it receives wake-up packets, it turns on the main transceiver.

One classic example of this is the video doorbell, which needs high data (because it is video); if it is battery-operated, power needs to be saved. A video doorbell needs to continuously sniff the air for online commands to operate the lock and that drains power. So, instead of keeping the main WLAN receiver on all the time (which is actually meant for high data rate), just for sniffing purpose, the wireless wake-up receiver can check for wake-up packets and then turn on the main WLAN receiver. This saves a considerable amount of power, which helps the IoT get high data rate, low latency as well as low power.

The main ways in which 802.11ba enables IoT are listed below.

- Battery life can be increased by ten times because the wake-up receiver is a simple receiver. It is a sniffing radio and consumes less than ten per cent of the actual WLAN receiver power.
- The wake-up receiver can always be kept on. Thus, its response time can be in milliseconds.
- It can be always connected. As an example, let us take the case of sensors on the move, like the mobile application with sensors moving around. If there is a disconnection from the access point (AP), instead of keeping the WLAN receiver on to sniff to see whether one is within AP

Comparison between 802.11ac and 802.11ax

	802.11ac	802.11ax
Spectrum	5GHz	2.4 and 5GHz
Technology	OFDM	OFDMA
Bandwidth	20/40/80/80+80/160MHz	20/40/80/80+80/160MHz
Highest modulation	256 QAM	1024 QAM
Sub-carrier spacing	312.5kHz	78.125kHz
Symbol length	3.2μs	12.8μs
Guard interval	0.4/0.8μs	0.8/1.6/3.2μs
MU-MIMO	4 (DL)	8 (UL and DL)
Spatial streams	8	8
Data rates	433Mbps (80MHz 1SS)	600.4Mbps (80MHz 1SS)
Spectral efficiency	42.5bps/Hz	62.5bps/Hz

coverage, one can use the wake-up receiver to sniff for AP coverage.

This again saves considerable power.

There are future advances possible for 802.11ba. The wake-up packet consists of a legacy field and a data field. Data field can be used to selectively turn on devices to have addresses for specific devices and those devices with the wake-up packet. Other uses of the data packet are as follows:

- Indoor location scan, which can be implemented using data packets from neighbouring access points (APs) to figure out one's location.
- Roaming scan; when one moves out of AP coverage, WLAN receivers turn on a scan to check for new, available networks. This can be done on a wake-up receiver, which substantially reduces power consumption for the WLAN device.

Wi-Fi 6, also called 802.11ax, is the latest upgrade for WLAN. This is a substantial improvement over the previous standard, Wi-Fi 5 or 11ac. Wi-Fi 5 was mostly focussed on increasing data throughput in favourable conditions. Wi-Fi 6 tries to achieve the best possible average data output from WLAN. Though WLAN is capable of gigabits/sec kind of data rate, it is normally not possible. Data rate substantially drops when there are multiple connections to the same AP. Wi-Fi 6 is trying to address this issue.

There are two major changes between ac and ax, as given below.

- The technology has changed from orthogonal frequency-division multiplexing (OFDM) to orthogonal frequency division multiple access (OFDMA).
- Sub-carrier data spacing (frequencies that carry data symbols) has reduced from 312KHz to 78KHz. 78KHz means that there are data bins/carriers closely packed together.

Usually, WLAN channel is 20MHz long, which can pack four times more data carriers using lower sub-carrier spacing.

Symbol length and sub-carrier frequency are interlinked, which means if frequency reduces, symbol length increases.

The advantage of having a longer symbol length is that one is more immune to delays of signal reception. So, in case of multiple signal reflections, each reflection is slightly delayed. When that happens, the longer the symbol length, the more robust it is, without inter-symbol interference.

Longer symbol length helps achieve a bigger guard interval, which is 3.2 μs versus 0.8 μs. A bigger guard interval safeguards from multiple reflections and helps get more reliable data. This improves average throughput in dense environments when connections to AP increase.